

UDAY KUMAR GUDUGUNTLA

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Professional Summary

- Python Developer and NLP Engineer with experience building AI-powered customer support solutions in a SaaS environment. Skilled in developing agentic AI systems, chatbots, voice bots, RAG pipelines, MCP servers, and semantic data layers that enable LLMs to understand and interact with enterprise data effectively. Experienced in FastAPI-based backend development, conversational AI, prompt engineering, and scalable AI integrations focused on solving real-world business automation challenges.

Experience

NLP Developer

May 2025 – Present

Worktual Innovations Pvt. Ltd.

Hyderabad, Telangana

- Developed AI-powered chatbots and voice bots for customer support automation, improving conversational experiences across SaaS products.
- Worked within the NLP Research & Development team to evaluate emerging AI technologies and integrate production-ready solutions into enterprise platforms.
- Designed and enhanced agentic AI workflows using context engineering, session memory, and long-term memory strategies to improve response accuracy and conversational continuity.
- Contributed to Retrieval-Augmented Generation (RAG) systems and knowledge retrieval pipelines, reducing inaccurate responses and improving information grounding.
- Built and maintained FastAPI-based backend services for AI applications, enabling scalable integration between LLMs, enterprise systems, and customer-facing products.
- Participated in the development of MCP (Model Context Protocol) servers and tool-integration frameworks to extend AI agent capabilities.
- Contributed to a Hub-and-Spoke AI architecture that unifies multiple products through a common intelligence layer, enabling business users to access customer insights and cross-platform analytics.
- Collaborated on the design of a semantic data layer backed by customer data platforms (CDPs), allowing LLMs to understand business entities and relationships through semantic models rather than direct database access.

Projects

Multi-Agent Customer Support Orchestration Platform

- Consolidated 7+ AI-powered chatbots and voice bots into a unified orchestration layer capable of routing user requests to the appropriate agent and toolset.
- Designed a centralized request handling framework that dynamically selects tools, workflows, and business functions based on user intent.
- Improved maintainability and scalability by reducing duplicated logic across multiple conversational AI systems.

Enterprise Semantic Layer for AI Applications

- Contributed to the development of a semantic layer that enables LLMs to understand business entities, relationships, and customer data beyond traditional table-based database access.
- Worked with PostgreSQL, Apache AGE, and graph-based modeling approaches to represent business knowledge and entity relationships.
- Improved contextual understanding for AI systems by leveraging ontology-driven data representations and semantic querying techniques.

MCP Server and Tool Management Framework

- Developed MCP (Model Context Protocol) server capabilities for exposing enterprise tools and services to AI applications.
- Designed a tool access architecture that allows individual bots to receive only the tools required for their specific business workflows.
- Supported scalable integration between conversational agents, backend services, and enterprise systems.

Technical Skills

Programming Languages - Python, SQL, Java

AI & NLP - Natural Language Processing (NLP), Large Language Models (LLMs), Agentic AI, Retrieval-Augmented Generation (RAG), Prompt Engineering, Context Engineering, Conversational AI

Backend Development - FastAPI, REST APIs

Databases - MySQL, PostgreSQL, Neo4j

Semantic & Graph Technologies - Semantic Layer Design, Apache AGE, Graph Databases, Knowledge Representation

AI Infrastructure - MCP (Model Context Protocol), Tool Integration, Semantic Search, Vector Search

Tools - Git, GitHub, pgAdmin

Education

MCA — Narayana Engineering College, Nellore	CGPA: 8.8 — 2023 – 2025
B.Sc in Statistics — Vivekananda Degree College, Nellore	CGPA: 9.2 — 2020 – 2023
12th (BIEAP) — Sri Chaitanya Jr College, Guntur	Percentage: 84.0% — 2016 – 2018
10th (SSC) — Kenedy High School, Guntur	CGPA: 7.5 — 2015 – 2016

Certifications & Achievements

Anthropic Academy – Introduction to Model Context Protocol (MCP)

Top 40 Finalist – CVCORP Hackathon (Ranked among 4,000+ participants)

1st Prize – Code Change, TechChroma National-Level Technical Symposium

2nd Prize – Reverse Coding, IEEE SB NBKRIST

1st Prize – DevFista Hackathon by Nexzera and SkillBridge

Leadership & Community Engagement

- Technical Trainer (CodeTantra)**
- Trained 1,000+ students in Python, C Programming, and Data Structures across multiple engineering institutions.
 - Developed learning materials and coding assessments adopted across partner colleges.
 - Conducted placement-focused training sessions emphasizing problem-solving and coding proficiency.

- Technical Event Leadership**
- Organized a Web Development Workshop for 70+ students covering full-stack development fundamentals.
 - Led the organization of TechChroma, a national-level technical symposium for MCA students.

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